



HOUSEHOLD AI / SOVEREIGNTY BRIEF

Your home should own the rules, not every model.

Sovereign AI at human scale: an architecture for enforceable household control, portable intelligence, bounded action, continuity, and evidence.

The argument

Sovereignty is an executable property, not a residency promise. The household should control the context, authority, action, evidence, and exit path while models remain replaceable.

Research synthesis and design hypothesis / July 2026. This brief does not claim an implemented runtime or original empirical validation.

Sovereignty is decision capacity.

Leading 2025-2026 work treats AI sovereignty as the capacity to make deliberate decisions across infrastructure, data, models, applications, talent, operations, security, and supply dependencies - not as a binary claim that everything is locally owned. [1][2][3]

Full-stack self-sufficiency is unrealistic for most nations and even less practical for a household. The credible alternative is managed interdependence: map dependencies, diversify suppliers, preserve portability, and keep decision rights and continuity under the user's control. [3]

At household scale, sovereignty becomes concrete: who may use which context, where a capability may run, which provider may supply it, what action it may take, and what evidence it must leave behind. Local execution is one instrument of control, not its definition.

Field-level control	Household question	Executable proof
Data and AI	Who may use which context, for what purpose?	Scoped grants, field rules, provenance, deletion and access evidence
Operational	Who can run, support, recover, and change the system?	Safe degraded mode, cached policy, tested recovery and provider exit
Technology and supply	Can models, devices, and services be substituted?	Stable capability contracts, open interfaces, export and regression tests
Security and governance	Who may approve or cause a real-world change?	Typed actions, contextual guards, human checkpoints, receipts

The household should own the rules, not every model.

The control point moves above the model.

1. Sovereignty is a control system.	Residency and local inference can help, but they do not by themselves decide who holds keys, sets policy, approves actions, sees evidence, or can replace a supplier.
2. The home needs a shared map, not one universal memory.	People, roles, accounts, devices, documents, tasks, and permissions need explicit relationships. Personal, child, guest, shared, and employer knowledge remain separate by default. Task-specific capability matters more than a monolithic AI label. [4]
3. The model is a replaceable supplier.	The task boundary, context contract, quality bar, approval rule, and recovery path should remain stable when the underlying model or execution service changes.
4. Authority belongs to typed actions.	Fluent output is not permission. Explain, draft, pay, order, unlock, share, and delete are distinct actions with distinct identity, state, policy, approval, and evidence requirements.
5. Exit, continuity, and evidence are the test.	A sovereign household can keep safe functions working through disruption, substitute approved suppliers, export its map and policies, and show what context, route, decision, and change occurred.

Shared meaning. Separate knowledge.

The architecture needs a policy-filtered map of household reality, not a centralized copy of every private record. It stores identities, relationships, permissions, pointers, minimum shared state, and evidence while leaving private content in its authorized scope where possible.

People and roles Adult, guardian, child, guest, caregiver, delegated operator, employer identity.	Places and devices Rooms, phones, hubs, PCs, appliances, vehicles, engines, trust and resource state.
Accounts, assets, obligations Bills, subscriptions, warranties, appointments, inventory, vehicles, service relationships.	Knowledge scopes Private, shared, guardian, child, guest, and employer records with provenance and retention.
Capabilities Rules, specialists, generalists, remote providers, validators, tools, versions, and evaluated scope.	Policies Purpose, consent, context grants, authority tier, residency, budget, deadline, escalation.
Typed actions Parameters, preconditions, permissions, effects, reversibility, approval, and evidence.	Evidence and outcomes Routes, grants, validations, approvals, receipts, postconditions, errors, and undo windows.

The graph is not a digital twin.

Household state can be incomplete, stale, disputed, or inferred. Every assertion needs source, owner, freshness, sensitivity, and policy. Missing information is not proof that something is false.

Open, machine-readable device descriptions, typed actions, and provenance models offer useful standards precedents for this layer. [8][9]

Every request crosses five gates.

The household states a job. The system resolves the applicable people, assets, relationships, context, and current state; enforces authority before optimization; chooses and verifies a capable implementation; then acts, asks, hands off, or stops.



Context grant Minimum authorized data references, purpose, fields, expiry, retention, and policy version.	Capability lease Approved function, model or rule; allowed device; tools; resource fit; timeout; fallback.
Action proposal Typed target, parameters, expected state, risk, reversibility, approval, and idempotency.	Action receipt Requester, context, route, versions, policy decision, validation, approval, effects, and outcome.

The model may propose. The action contract decides whether anything happens.

Consequential changes run only through explicit action interfaces. Identity, purpose, target state, parameters, policy, and approval are rechecked at commit time. An intelligible denial, abstention, deferral, or handoff is a valid outcome. [5][6][7][20]

Useful because authority is legible.

Catch a bill increase Resolve the account owner, tariff, prior statements, requester role, and evidence sources before a model sees the task. Boundary: Explain and draft. Do not pay, cancel, or close an account without fresh approval.	Help a child learn Resolve the child, assignment, guardian rules, and only the school context approved for the learning task. Boundary: Teach and check understanding. Do not complete assessed work or impersonate the student.
Troubleshoot an appliance Resolve the exact appliance, warranty, service history, and manual before a specialist proposes a check. Boundary: Guide low-risk, manual-supported steps. Stop for hazards or warranty-voiding work.	Plan groceries Combine only authorized inventory, shared lists, allergies, and preferences without flattening private profiles. Boundary: Suggest and update the shared list. Do not merge private data or check out without approval.

Household activity surveys can identify recurring task families, but they are demand evidence - not proof that a task should be automated. [10] These examples are design tests, not claims of empirical performance.

The operating layer becomes strategic.

Builders	Device makers	Investors
Build the household operating graph. Represent people, roles, assets, context scopes, capabilities, policies, actions, and evidence. Bind models only after the outcome and authority contract exists.	Expose capability, trust, and action. Publish typed device capabilities, action interfaces, trust boundaries, and measured resource envelopes under real operating constraints.	Test whether the moat respects exit. Governed household state, permission relationships, typed actions, and outcome evidence matter only if they remain portable and auditable.

Five tests, plus economic reality

Authority Can policy permit, constrain, approve, or deny the action?	Context Do personal, shared, child, guest, and employer scopes stay separate?
Substitutability Can a supplier change without losing capability, policy, or history?	Continuity Is there safe degraded operation, recovery, export, and exit?
Evidence Can the household see what context, route, decision, and change occurred?	Economic reality Do benefits survive full operating, validation, and failure costs?

Economic rule: optimize for cost per policy-compliant successful outcome, not token price. Count storage, updates, energy, cold starts, device interference, validation, routing errors, human review, outage recovery, portability work, and failure loss. Model-routing research suggests potential cost-quality gains; the household thesis adds authority, device, continuity, and evidence constraints. [13][14][15]

The integration is the test.

Fog and service-routing work already address heterogeneous placement. Model routers trade quality and price. Consumer frameworks and smart-home pipelines already choose runtimes and compose services. Privacy systems use attestation and minimized cloud processing. Governance and web standards already provide lifecycle controls, typed device interactions, and provenance. [6][8][9][11][12][13][14][15][16][17][18][19]

The defensible hypothesis is narrower: a portable household operating layer that joins task-specific capability, a distributed household graph, scoped identity and knowledge, heterogeneous placement, typed actions, evaluation gates, evidence, and safe substitution. This is a synthesis to test, not a claim that hybrid inference, policy engines, semantic graphs, or bounded agents were invented here.

Policy before optimization No cheaper, faster, or more capable route may weaken context, identity, jurisdiction, or authority.	Competence before placement No model or bundle serves a task family until the complete workflow passes its evaluation gate.
No direct model writes Consequential changes use typed actions with admission, current-state checks, approval, and receipts.	Feedback cannot rewrite policy Verified outcomes may propose changes; evaluation, approval, canary, and rollback govern adoption.

The honest falsification test

The thesis loses if one compact generalist plus tools delivers the same outcomes, control, continuity, portability, and full cost with less complexity. It also loses if graph maintenance becomes too costly or stale, competence boundaries are unreliable, device fragmentation overwhelms gains, or supplier substitution is performative.

Judgment: useful synthesis but not yet novel. Distinct technical utility requires an open reference implementation and benchmark showing that these contracts outperform simpler baselines without weakening household authority.

Evidence behind the brief

Sources prioritize public frameworks, official standards, first-party technical documentation, and peer-reviewed or clearly labeled preprint research. Accessed July 2026.

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- [20] Yonatan Geifman and Ran El-Yaniv. Selective Classification for Deep Neural Networks. NeurIPS 2017. <https://papers.nips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html>

Scope note: this brief proposes an architecture and evaluation agenda. It does not claim that the operating graph is complete, that typed actions make arbitrary agents safe, that external effects are always reversible, or that model substitution works without system-level regression testing.